

CS:5980

Foundations of Embedded Systems

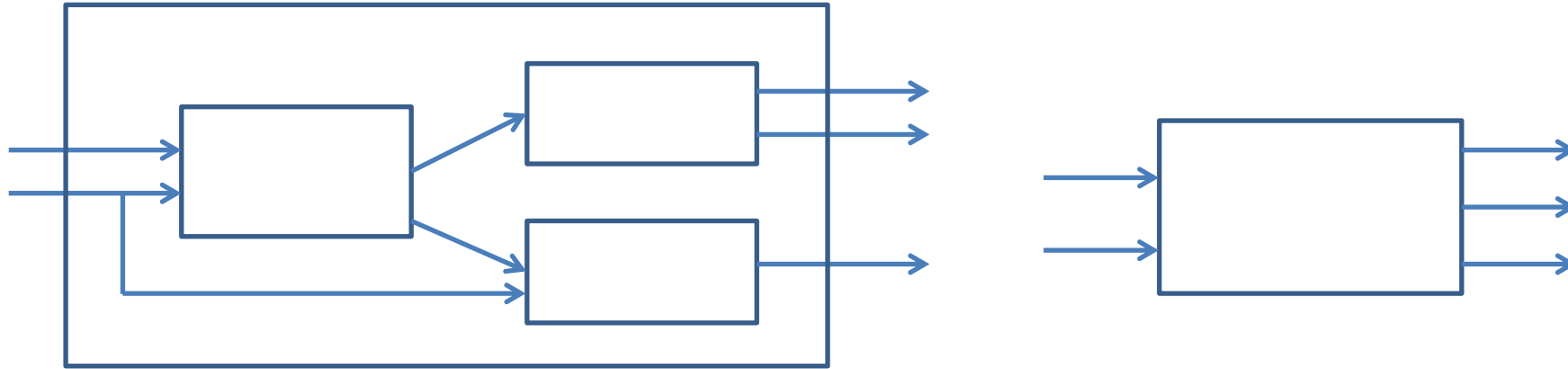
The Synchronous Model

Part I

Copyright 2014-26, Rajeev Alur and Cesare Tinelli.

Created by Cesare Tinelli at the University of Iowa from notes originally developed by Rajeev Alur at the University of Pennsylvania. These notes are copyrighted materials and may not be used in other course settings outside of the University of Iowa in their current form or modified form without the express written permission of one of the copyright holders. During this course, students are prohibited from selling notes to or being paid for taking notes by any person or commercial firm without the express written permission of one of the copyright holders.

Model-Based Design



Block Diagrams

- Widely used in industrial design
- Tools: Simulink, Modelica, LabView, RationalRose, ...

Key question: what is the execution semantics?

- What is a base component?
- How do we compose components to form complex components?

Functional vs. Reactive Computation

Functional model of computation (classical one):

- Given inputs, a program produces outputs
- Desired functionality described by a mathematical function
- **Example:** sorting of names; shortest paths in a weighted graph
- Theory of computation provides foundation
- **Canonical model:** Turing machines



Reactive model of computation:

- System continually interacts with its environment via inputs and outputs
- Desired behavior described by sequences of observed input/output interactions
- **Example:** cruise controller in a car

Sequential vs. Concurrent Computation

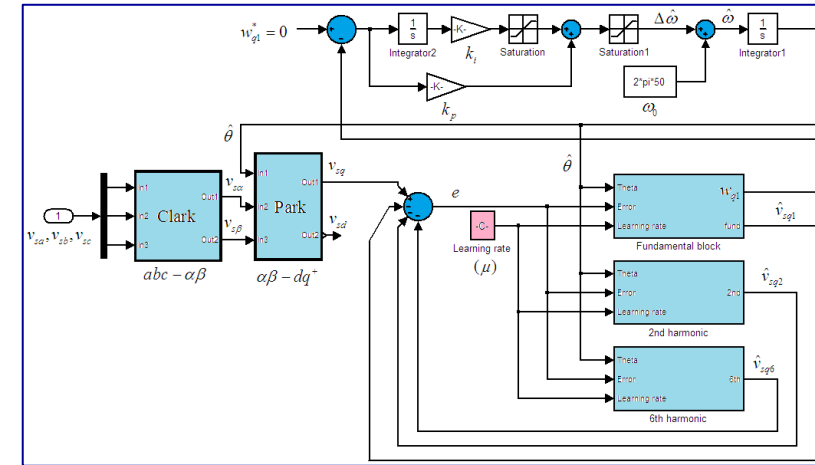
Sequential model of computation (classical one):

- A computation is a sequence of instructions executed one at a time
- Well understood and canonical model: Turing machines

Concurrent model of computation

- Multiple components/processes exchanging information and evolving concurrently
- Logical vs. physical concurrency
- Broad range of formal models for concurrent computation
- **Key distinction:** synchronous vs. asynchronous

Synchronous Models



All components execute in a sequence of (logical) rounds, in lock-step

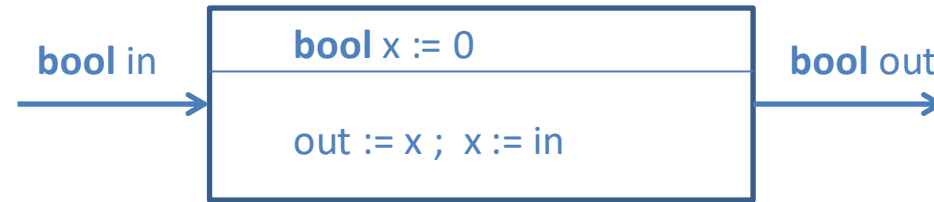
Example: Component blocks in digital hardware circuit

- Clock drives all components in a synchronized manner

Key idea in synchronous languages:

- Design system based on such synchronous round-based computation
- **Benefit:** design is simpler (why?)
- **Challenge:** ensure synchronous execution even if implementation platform is not single-chip hardware

First Example: Delay



Input variable: **in** of Boolean type ($\{0,1\}$)

Output variable: **out** of Boolean type

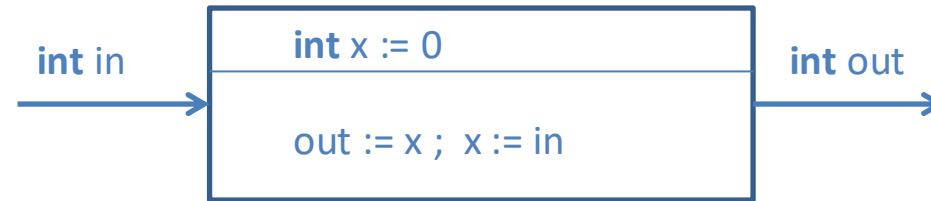
State variable: **x** of Boolean type

Initialization (of state variables): **x := 0**

Execution: in each round, in response to the current input,

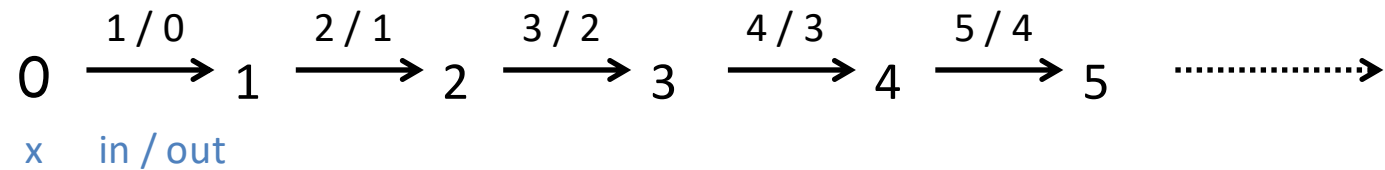
- produce output (**out := x**) and
- update state (**x := in**)

Int Delay: Round-based Execution

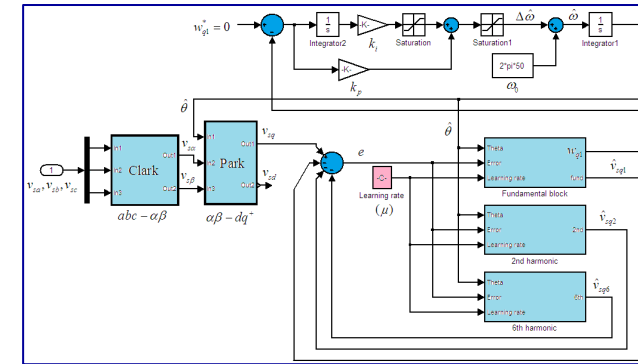


1. Initialize state x to 0
2. Repeatedly execute rounds. In each round:
 - a. Read current value of the input variable in
 - b. Execute the update code to produce output out
 - c. Change state as needed

Example execution:



Synchrony Hypothesis



Assumption: Time needed to execute the update code is negligible compared to delay between successive input arrivals

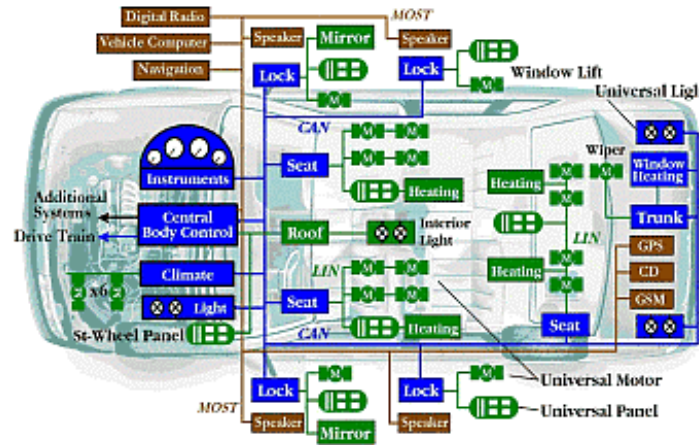
Logical abstraction:

- Execution of update code takes zero time
- Reception of inputs and production of outputs occur simultaneously

Composition: when multiple components are composed, **all** execute synchronously and simultaneously

Implementation: must ensure that this design-time assumption is valid!

Components in an Automobile



Components need to communicate and coordinate over a **shared bus**

Design abstraction: Synchronous **time-triggered communication**

- Time is divided into slots
- In each slot, exactly one component sends a message over the bus

Example: CAN protocol implements time-triggered communication

Model Definition

Syntax: How to describe a component?

- Variable declarations, types, state updates, ...

Semantics: What does the description mean?

- Defined using mathematical concepts such as sets, functions, ...

Formal foundations: Semantics is defined precisely

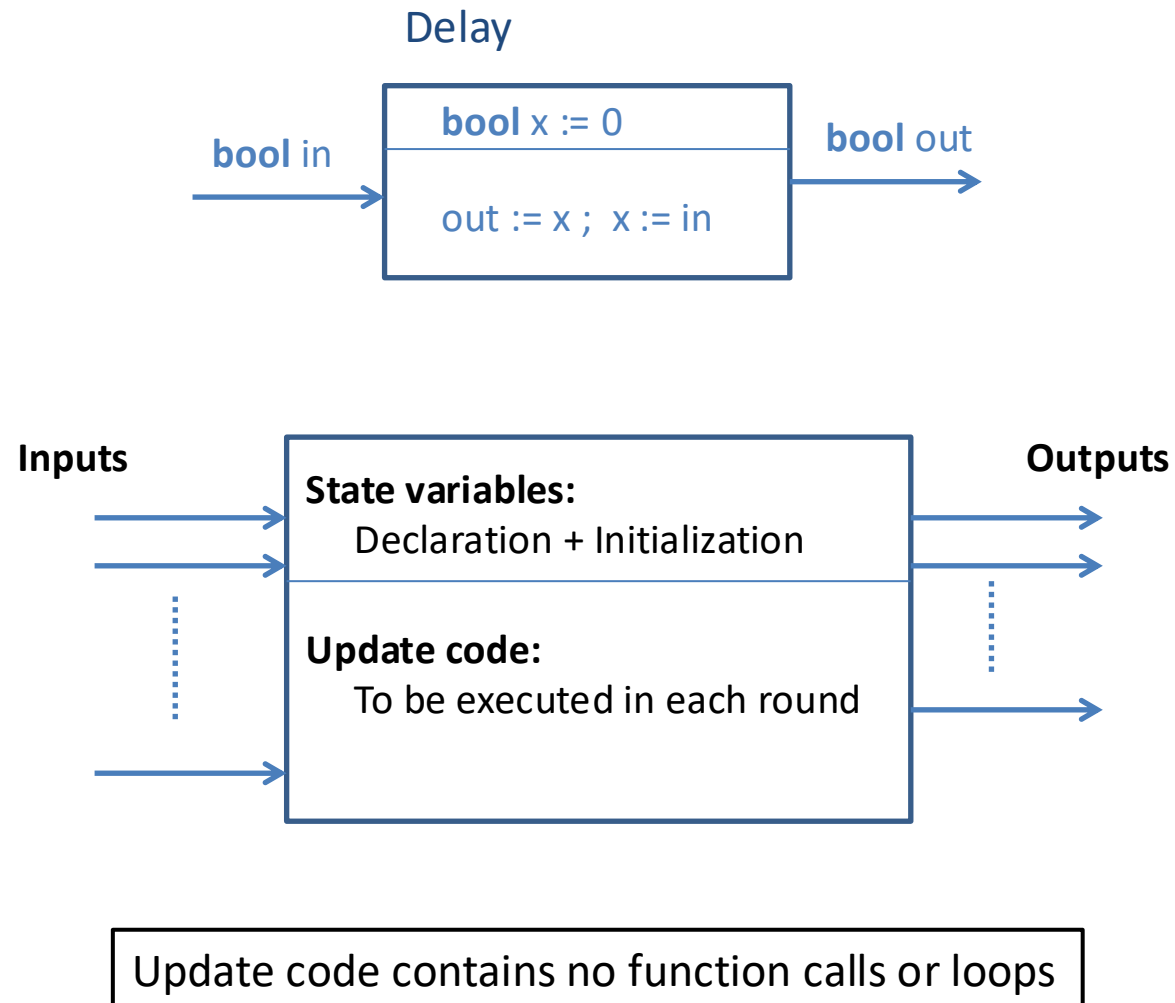
- Necessary for tools for analysis, compilation, verification, ...
- Defining formal semantics for a **real** language is challenging
- But concepts can be illustrated on a **toy** modeling language

Our Modeling Language

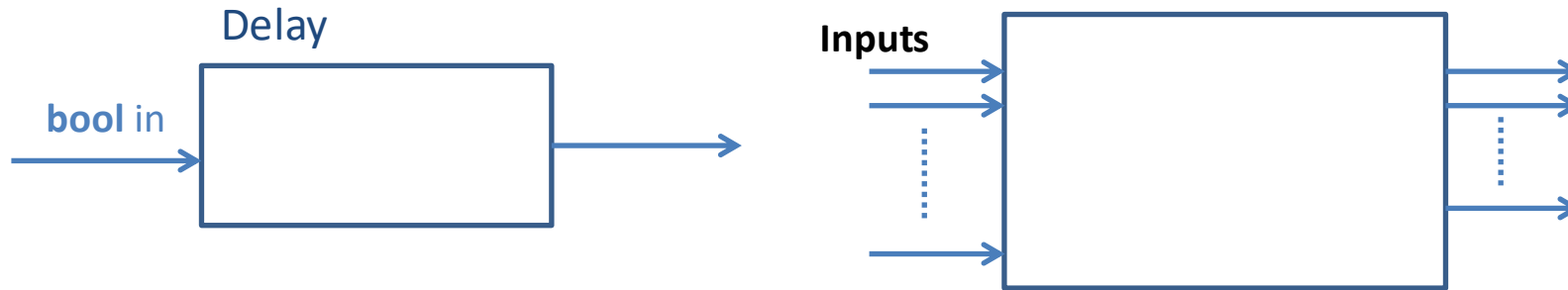
Synchronous Reactive Components

- Representative of many **academic** languages:
Signal, Lustre, Lucid, SMV, Lola, Lucent, ...
- Foundations of **industrial-strength** synchronous languages:
Esterel, SCADE, VHDL, Verilog, Stateflow, ...

Synchronous Reactive Component



SRC Definition — Inputs



Each component has a set I of input variables

- Variables have types. E.g., **bool**, **int**, **nat**, **real**, {on, off}, ...

Input: Assignment of all the input variables

- The set of possible inputs (input assignments) is Q_I

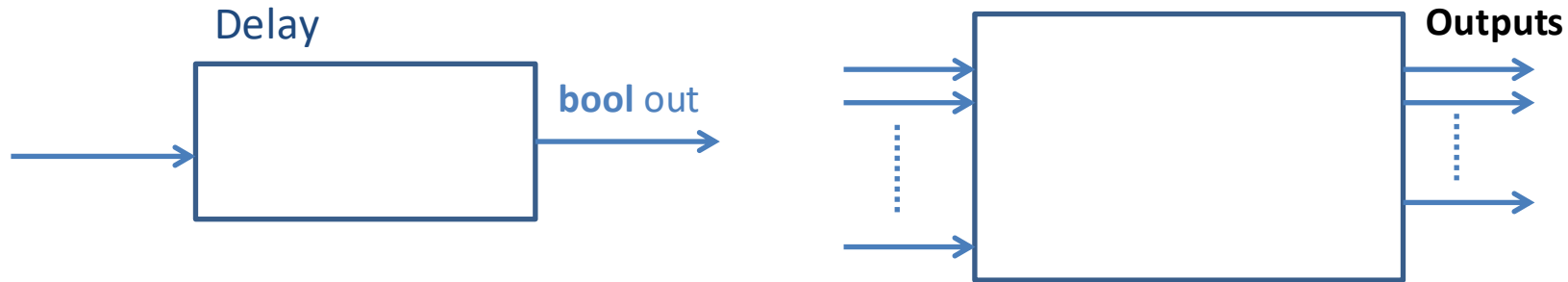
For **Delay**

- I contains a single variable in of type **bool**
- The set of inputs is $\{ \{ in := 0 \}, \{ in := 1 \} \}$

Example: I consists of two variables: **int** x , **bool** y

- Each input is a pair: (integer value for x and Boolean value for y)

SRC Definition — Outputs



Each component has a set O of typed output variables

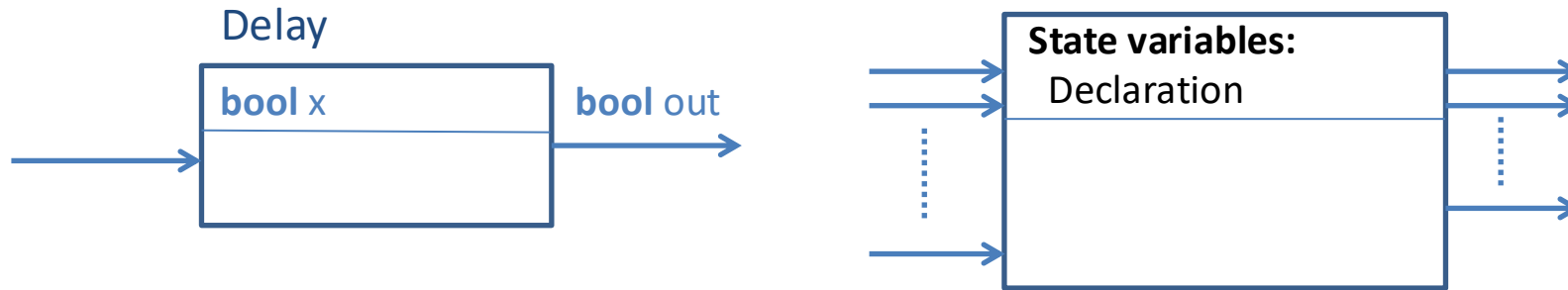
Output: Assignment of all the output variables

- The set of outputs (output assignments) is Q_o

For *Delay*

- O contains a single variable *out* of type **bool**
- The set of outputs is $\{ \{ \text{out} := 0 \}, \{ \text{out} := 1 \} \}$

SRC Definition — States



Each component has a set S of typed state variables

State: Assignment of all the state variables

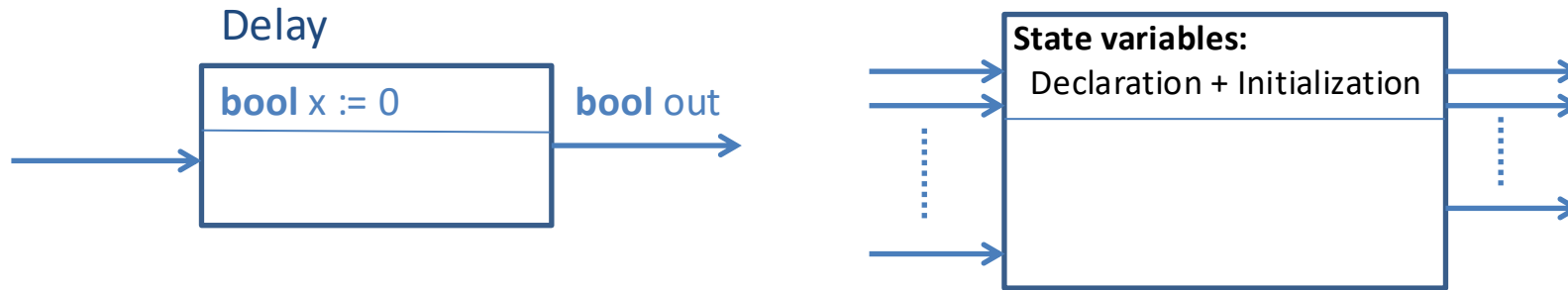
- The set of states is Q_S

For *Delay*

- S contains a single variable x of type **bool**
- The set of states is $\{ \{ x := 0 \}, \{ x := 1 \} \}$

State is internal and maintained across rounds

SRC Definition — Initialization



Initialization of state variables specified by **Init**

- A sequence of assignments to state variables

Semantics of initialization:

- The set **[Init]** of initial states, which is a subset of Q

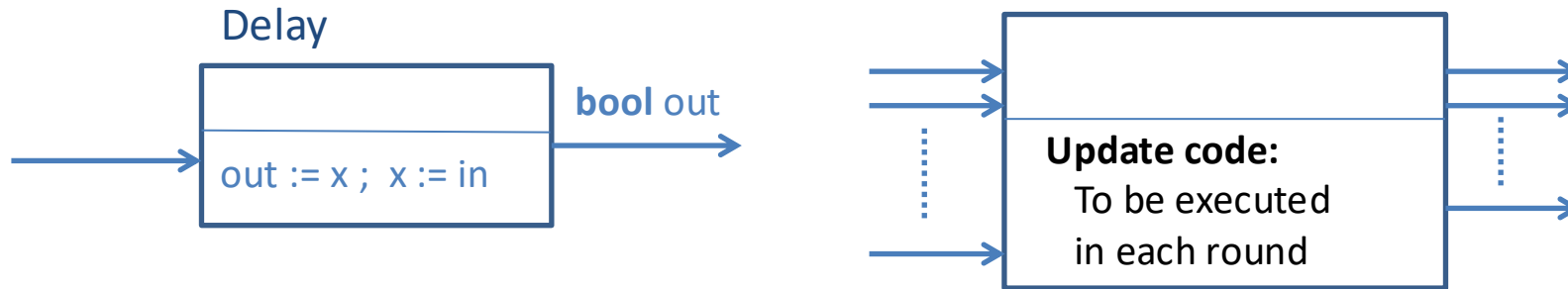
For **Delay**

- **Init** is given by the code fragment $x := 0$
- The set **[Init]** of initial states is $\{ \{ x := 0 \} \}$

Components can have multiple (alternative) initial states

- Example: **bool x := choose {0, 1}**

SRC Definition — Reactions



Execution in each round given by code fragment **React**

- Sequence of assignments and conditionals that assign output variables and update state variables

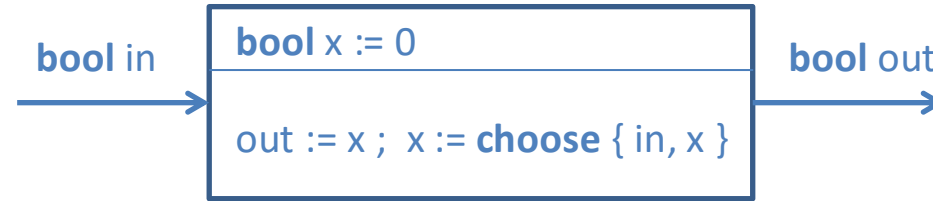
Semantics of update:

- The set **[React]** of reactions, where each reaction is of the form:
(old) state – input / output $\rightarrow$ (new) state
- **[React]** is a subset of $Q_S \times Q_I \times Q_O \times Q_S$

For **Delay**:

- **React** is defined by the code fragment `out := x ; x := in`
- 4 possible reactions: $0 - 0/0 \rightarrow 0$, $0 - 1/0 \rightarrow 1$, $1 - 0/1 \rightarrow 0$, $1 - 1/1 \rightarrow 1$

Multiple Reactions



During update, x is either updated to input in or is left unchanged

- Models the possibility that an input may be **lost or ignored**

Nondeterministic reactions:

- Given (old) state and input, output/new state need not be unique
- The set **[React]** of reactions now consists of

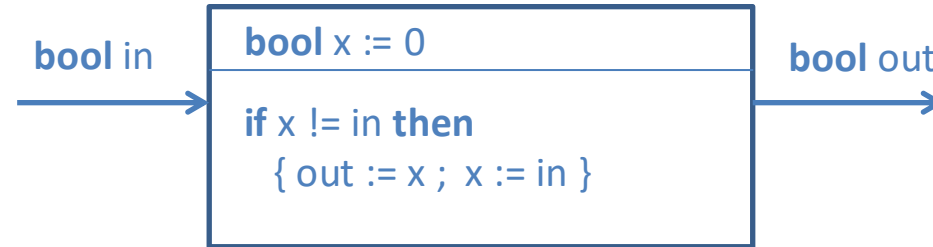
$0 - 0/0 \rightarrow 0$

$0 - 1/0 \rightarrow 1, \quad 0 - 1/0 \rightarrow 0$

$1 - 0/1 \rightarrow 0, \quad 1 - 0/1 \rightarrow 1$

$1 - 1/1 \rightarrow 1$

Selective Reactions



A component may not accept all inputs in all states

- **Motivation:** *blocking* communication

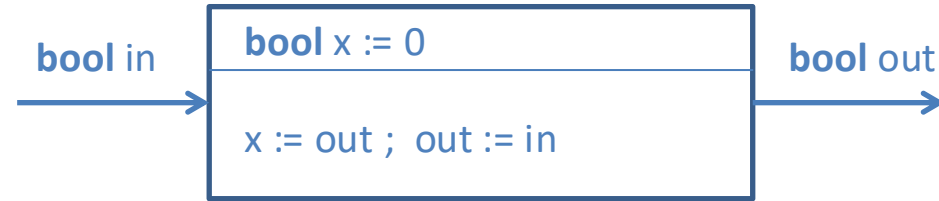
Possible set of reactions in certain state/input combinations may be empty

- The set `[React]` of reactions now consists of

$0 - 1/0 \rightarrow 1$

$1 - 0/1 \rightarrow 0$

Syntax Errors



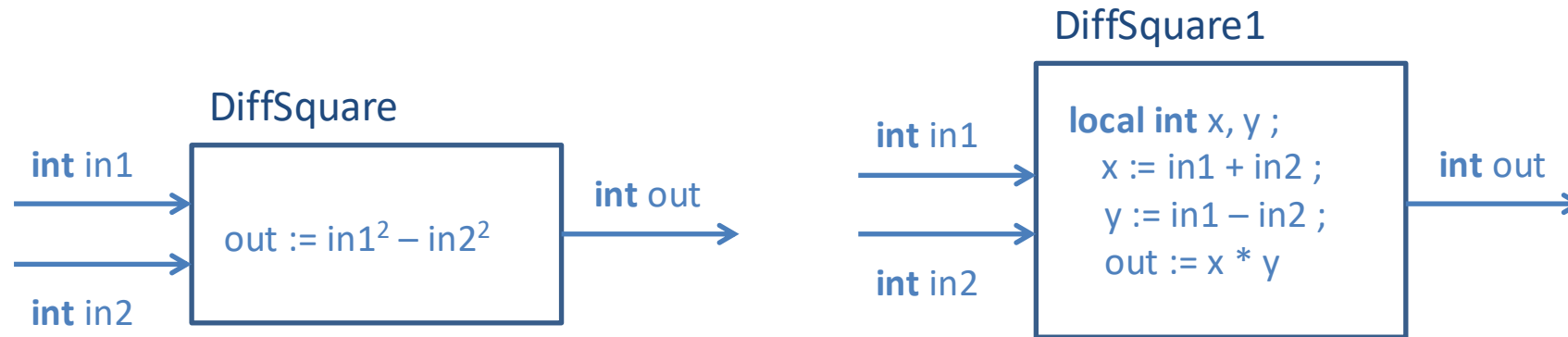
Update code is expected to satisfy a number of requirements:

- Types of variables and expressions should match
- **Output** variables must first be **written before** being **read**
- **Output** variables must be explicitly **assigned a value**

Otherwise, no reaction possible

- In above, the set **[React]** of reactions is empty

Semantic Equivalence



- ❑ These components have identical sets of reactions
- ❑ They are syntactically different but semantically equivalent
- ❑ Compiler can optimize code as long as semantics is preserved!

Synchronous Reactive Component Definition

- ❑ Set I of typed input variables gives set Q_I of inputs
- ❑ Set O of typed output variables: gives set Q_O of outputs
- ❑ Set S of typed state variables: gives set Q_S of states
- ❑ Initialization code $Init$: defines set $[Init]$ of initial states
- ❑ Reaction description $React$: defines set $[React]$ of reactions of the form $s - i/o \rightarrow t$, where s, t are states, i is an input, and o is an output

Synchronous languages **in practice**:

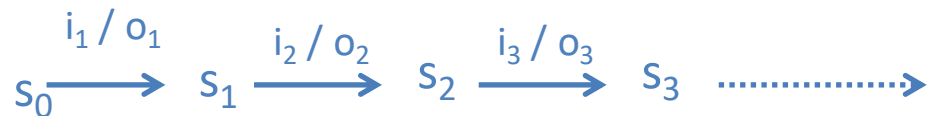
- Richer syntactic features to describe $React$
- Key to understanding: what happens in a single reaction?

Formal semantics: Necessary for development of tools!

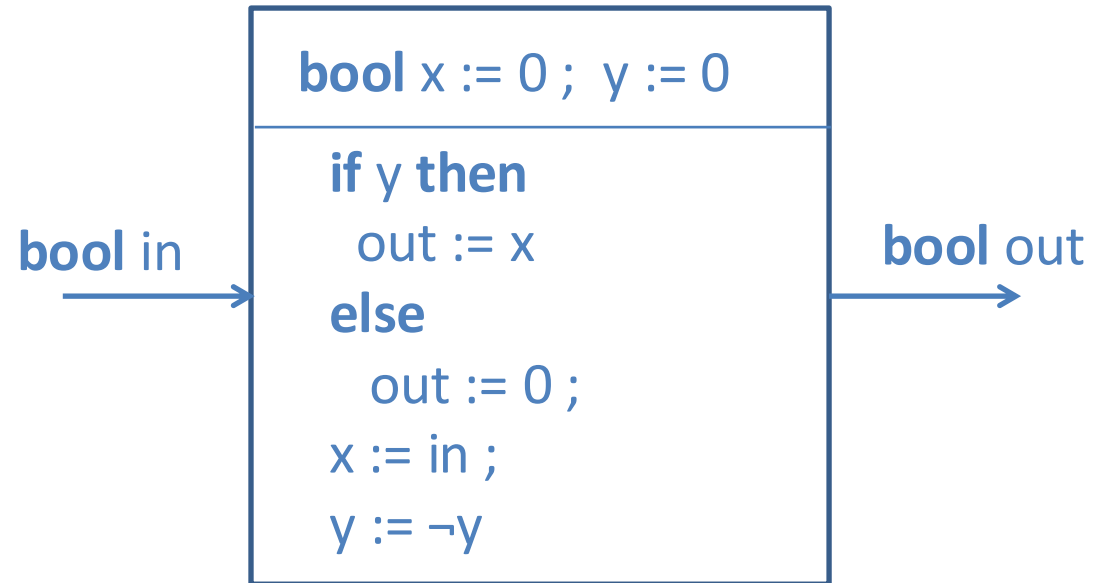
Definition of Executions

1. Given component $C = (I, O, S, \text{Init}, \text{React})$, what are its executions?
2. Initialize state to some state s_0 in $[\text{Init}]$
3. Repeatedly execute rounds. In each round $n = 1, 2, 3, \dots$
 - a. Choose an input value i_n in Q_I
 - b. Execute React to produce output o_n and change state to s_n
that is, $s_{n-1} - i_n / o_n \rightarrow s_n$ must be in $[\text{React}]$

Sample execution:

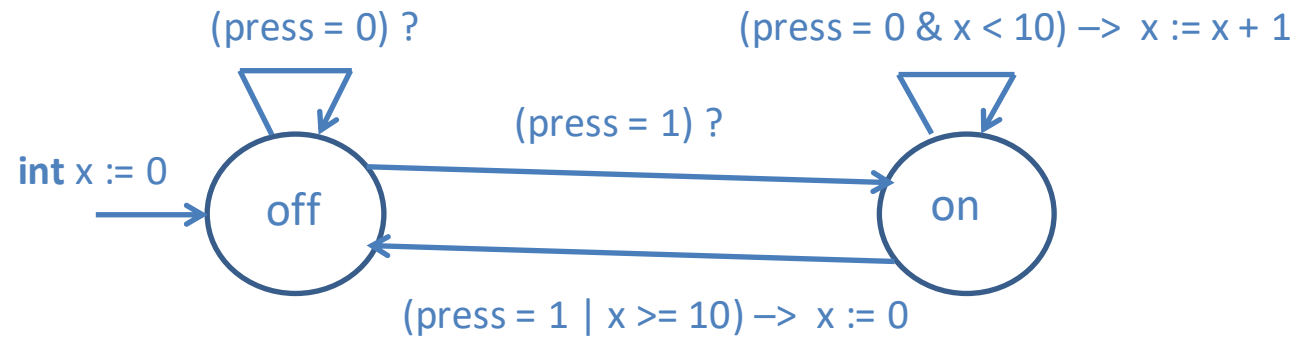


What does this component do ?



Extended State Machines

Input: **bool** press



mode is an implicit state variable ranging over $\{\text{on}, \text{off}\}$

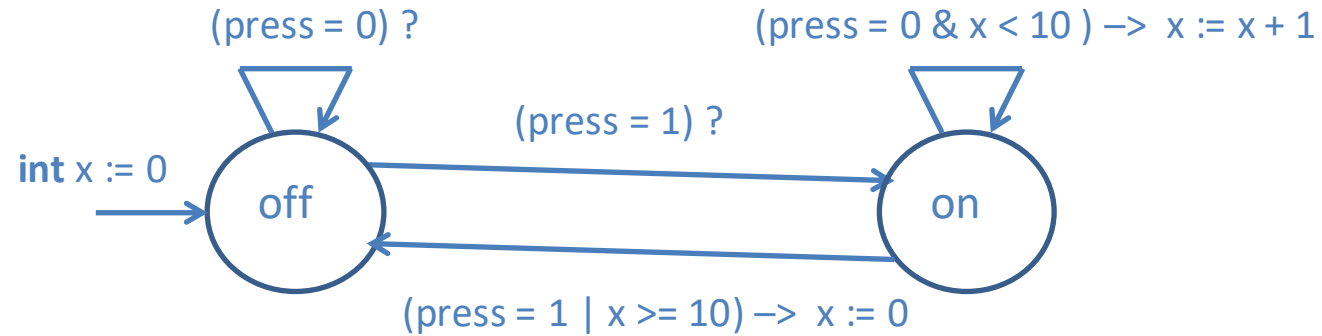
Reaction corresponds to executing a *mode-switch*

Example mode-switch: from **on** to **off** with

guard (**press** = 0 | **x** >= 10) and *update* **x** := 0

Executing ESMs: Switch

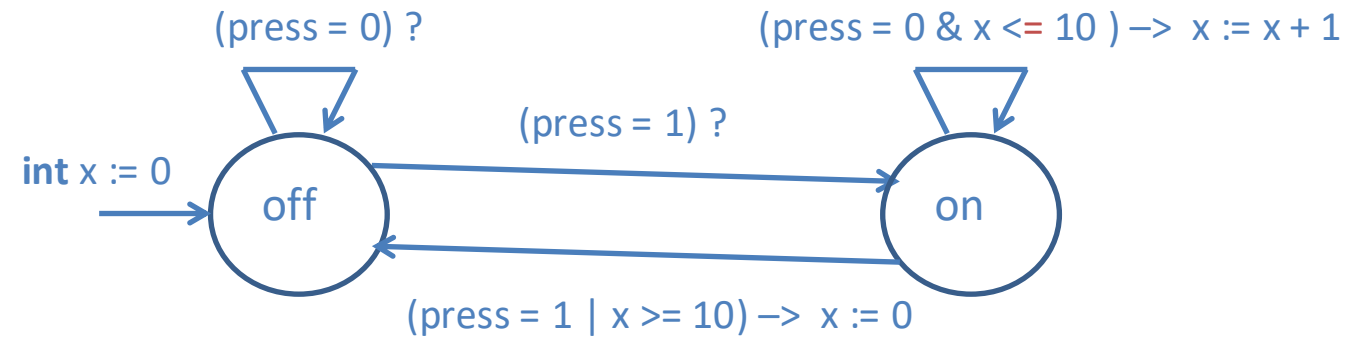
Input: **bool** press



- ❑ State of the component **Switch** assigns values to **mode** and **x**
- ❑ Initial state: **(off, 0)** (i.e., **{ mode := off, x := 0 }**)
- ❑ **Example execution:**
(off, 0) – 0 -> (off, 0) – 1 -> (on, 0) – 0 -> (on,1) – 0 -> (on,2) ... – 0 -> (on, 10) – 0 -> (off, 0)

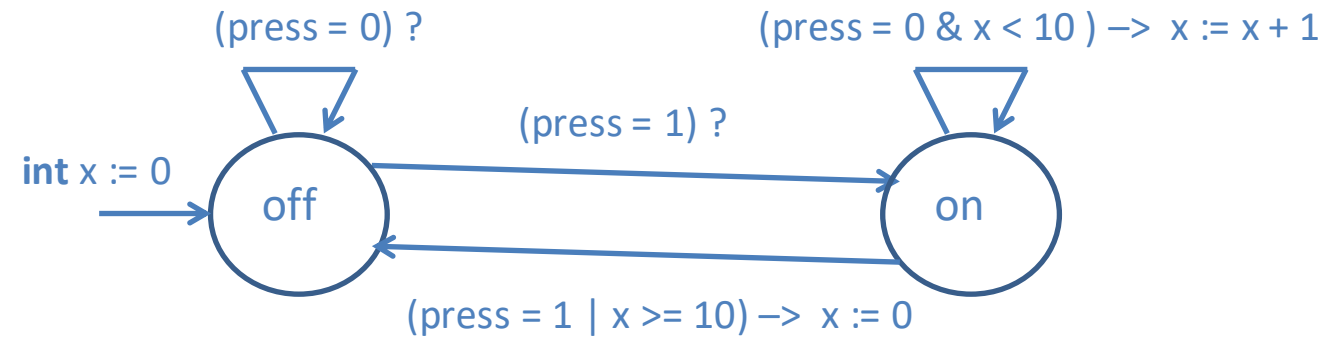
Modified Switch: What executions are possible?

Input: **bool** press



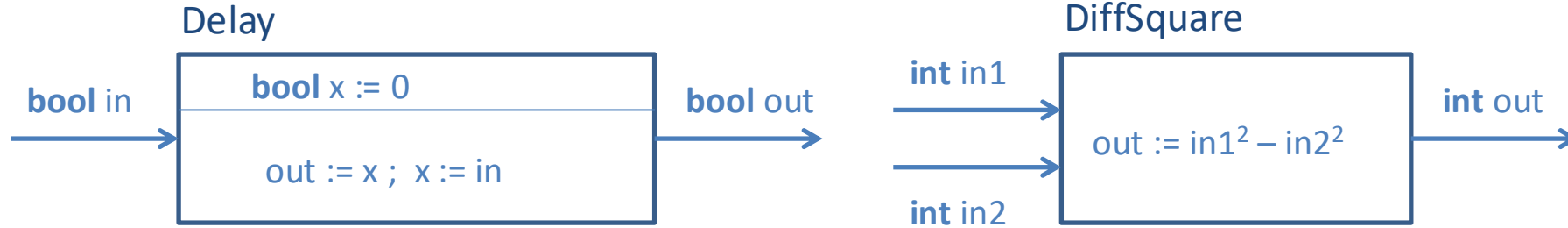
Exercise: ESM to SRC

Input: **bool** press



Rewrite this ESM as a synchronous reactive component

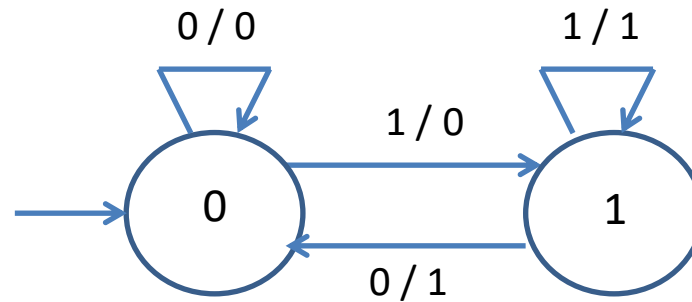
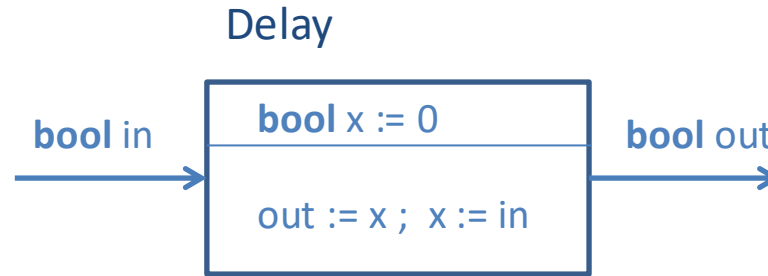
Finite-State Components



A component is *finite-state* if all its variables range over finite types

- Finite types: **bool**, enumerated types (e.g. **{on, off}**, **int[-5..5]**)
- **Delay** is finite-state, but **DiffSquare** is not

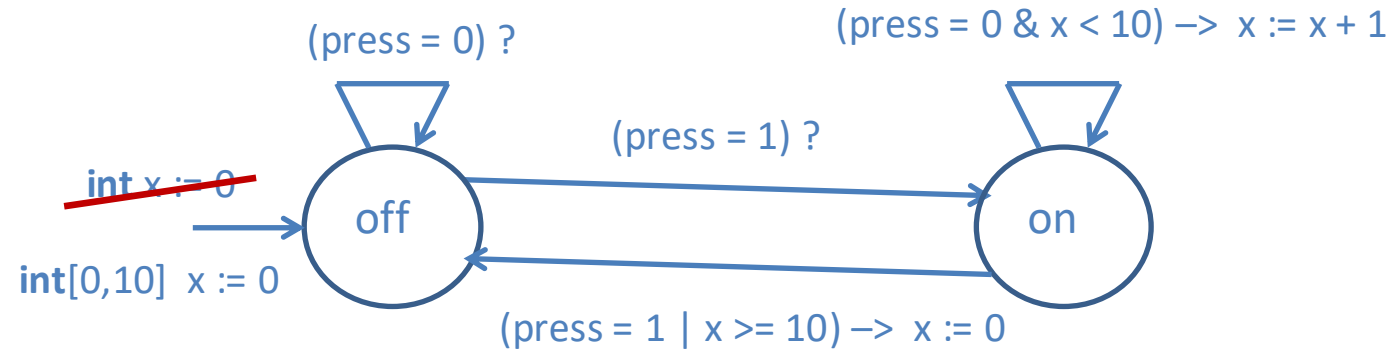
Mealy Machines (for finite-state components)



Finite-state components are amenable to exact, algorithmic analysis

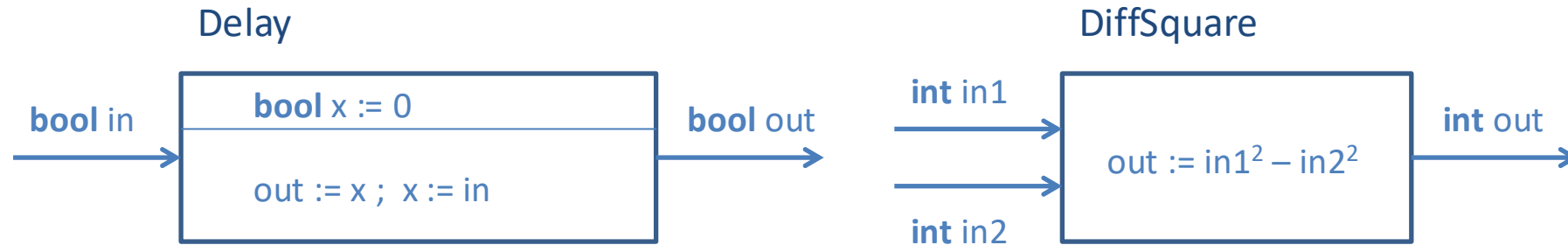
Switch: Is it finite-state?

Input: **bool** press



The system is **effectively** finite-state

Combinational Components



A component is *combinational* if it has **no state** variables

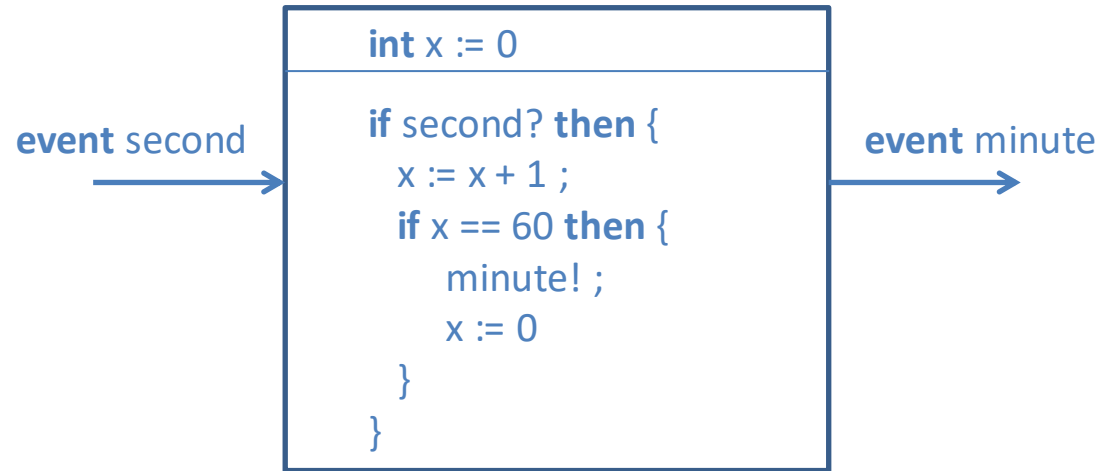
- `DiffSquare` is combinational, but `Delay` is not
- Hardware gates are combinational components

Events

- ❑ Input/output variable can be of type **event**
- ❑ **Motivation:** notion of clock can be different for different components
- ❑ An event can be absent, or present, in which case it has a value
 - **event** x means x ranges over $\{\text{present}, \perp\}$
 - **event(bool)** x means x ranges over $\{0, 1, \perp\}$
 - **event(nat)** x means x ranges over $\{\perp, 0, 1, 2, \dots\}$
- ❑ **Syntax:** $x?$ is a short form for the test $x \neq \perp$
- ❑ **Syntax:** $x!v$ is a short form for the assignment $x := v$
- ❑ **Syntax:** $x!$ is a short form for the assignment $x := \text{present}$ (for **event** x)
- ❑ **Event-based communication:**
 - If no value is assigned to an output event, then it is absent (by default)
 - Event-triggered component executes only in those rounds where input events are present (actual definition slightly more general, see textbook)

Second-To-Minute

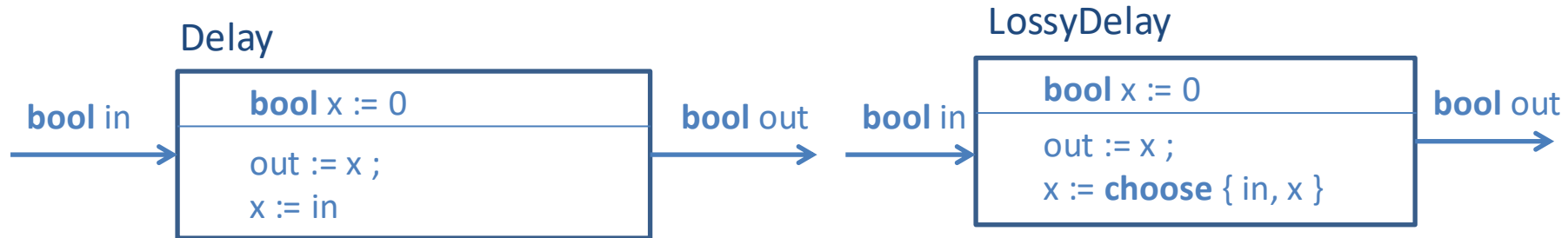
Requirement: Issue the output event every 60th time the input event is present



Event-Triggered Components

- No need to execute in a round where triggering input events absent
- See textbook for formal definition

Deterministic Components

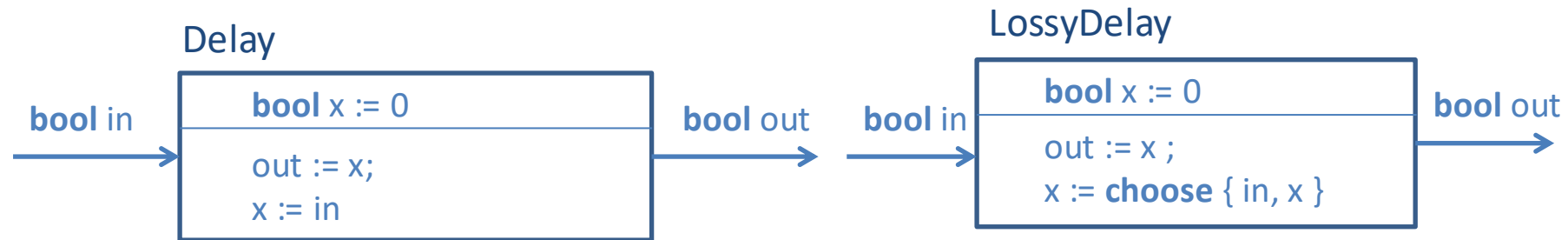


A component is *deterministic* if

1. it has a single initial state, and
2. for every state **s** and input **i**, there is a **unique** state **t** and output **o** such that **s – i/o → t** is a reaction

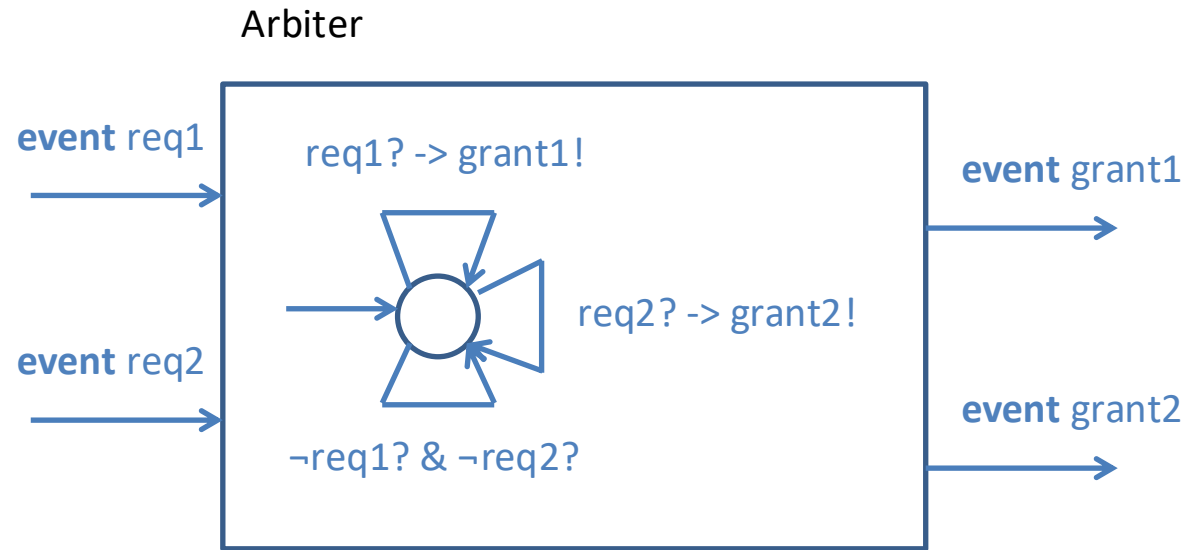
Delay is deterministic, but **LossyDelay** is not

Deterministic Components



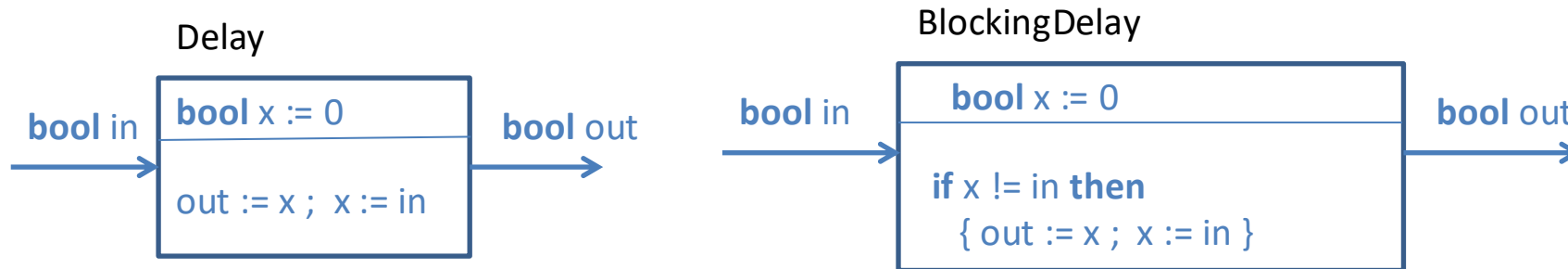
- ❑ **Determinism:** same sequence of inputs supplied,
same sequence of outputs observed (predictable, repeatable behavior)
- ❑ Nondeterminism is **useful** in modeling **uncertainty**, **partial knowledge** or **abstraction**
- ❑ Nondeterminism is **different** from **probabilistic** (or random) choice

Mixed Notation



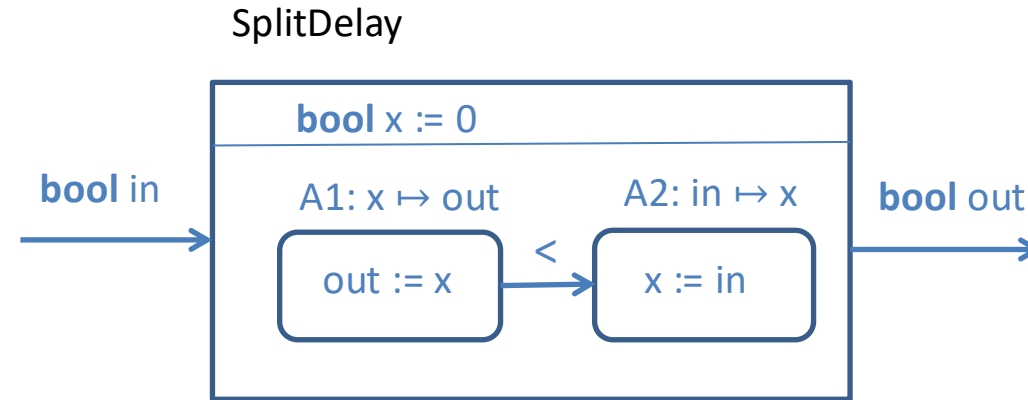
What does this component do?

Input Enabled Components



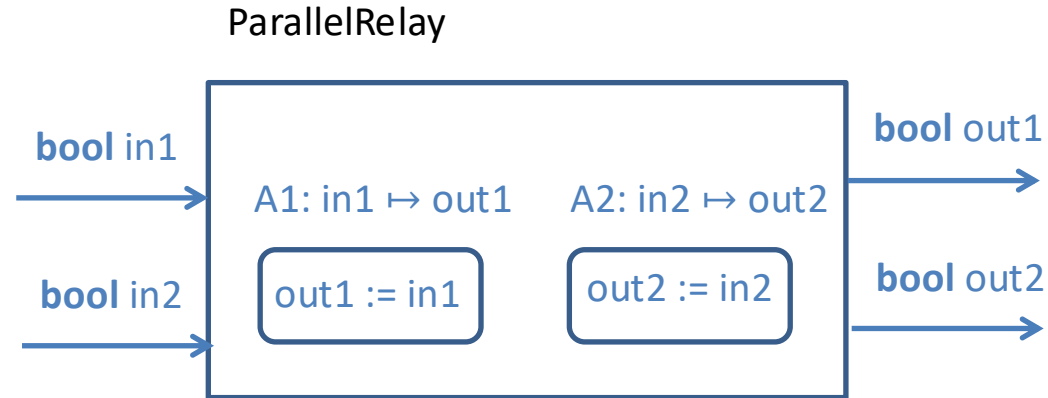
- ❑ A component is *input-enabled* if for every state s and input i , there exists a state t and an output o such that $s - i/o \rightarrow t$ is a reaction
 - Delay is input-enabled, but BlockingDelay is not
- ❑ Not input-enabled means component is making assumptions about the context in which it is going to be used
 - When rest of system is designed, must check that it indeed satisfies these assumptions

Splitting Reaction Code into Tasks



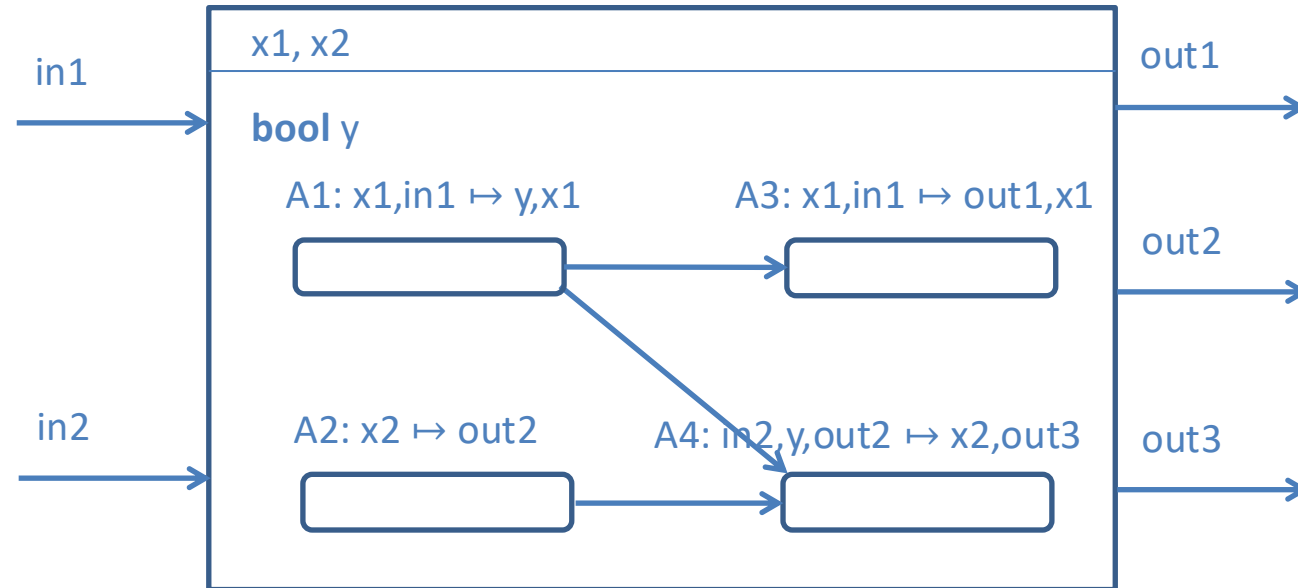
- ❑ **A1** and **A2** are tasks (atomic blocks of code)
 - Each task specifies variables it reads and writes
 - **A1** reads **x** and writes **out**
- ❑ **Task Graph**: vertices are tasks and edges denote precedence (<)
 - **A1** < **A2** means that **A1** should be executed before **A2**
 - Graph must be **acyclic**

Example Task Graph



- ❑ Tasks **A1** and **A2** are unordered
 - Possible *schedules* (linear ordering of tasks): **A1**, **A2** and **A2**, **A1**
 - All consistent schedules must give the **same** result
- ❑ I/O *await* dependencies: **out1** awaits **in1**, **out2** awaits **in2**

Example Task Graph



- ❑ What are possible schedules consistent with precedence constraints?
- ❑ What are I/O await dependencies?

Task Graphs: Definition

For a synchronous reactive component C with

- input vars I output vars O
- state vars S local vars L

the reaction description is given by

- a set of tasks, and
- precedence edges $<$ over these tasks

Each task A is specified by:

1. Read-set R
 - must be a subset of $I \cup S \cup O \cup L$
2. Write-set W
 - must be a subset of $O \cup S \cup L$
3. Update: code to write vars in W based on values of vars in R
 - $[Update]$ is a subset of $Q_R \times Q_W$

Requirements on Task Graph (1)

The precedence relation $<$ must be acyclic

- ❑ Notation: $A <^+ A'$ means that there is a path from task A to task A' in the task graph using precedence edges
- ❑ Relation $<^+$ is the *transitive closure* of the relation $<$
- ❑ *Task schedule*: Total ordering $A_1, A_2, \dots, A_n$ of all the tasks that is consistent with the precedence edges
 - If $A < A'$, then A must appear before A' in the ordering
 - Multiple schedules are possible
 - If $A <^+ A'$ then A must appear before A' in *every* schedule
- ❑ Acyclicity implies that there is at least one task schedule

Requirements on Task Graph (2)

Each output variable is in the write-set of exactly one task

- ❑ If output y is in write-set of task A , then as soon as A executes the output y is available to the rest of the system
- ❑ If task A writes output y , then y *awaits* an input variable x , written $y > x$, if
 - either the task A reads x
 - some another task A' such that $A' <^+ A$ reads x

Note: y awaits x means that y cannot be produced before x is supplied

Requirements on Task Graph (3)

Output/local variables are written before being read

- If an output or a local variable y is in the read-set of a task A , then y must be in the write-set of some task A' such that $A' <^+ A$

Requirements on Task Graph (4)

Tasks with a write conflict must be ordered

- ❑ There is a *write-conflict* between tasks A and A' if a variable written by A is read or written by A'
- ❑ If A and A' have a write-conflict, the result depends on whether A executes before A' or vice versa.
 - **Example:** A update is $x := x + 1$; A' update is $out := x$
- ❑ If tasks A and A' have a write-conflict, then they must be ordered: either $A <^+ A'$ or $A' <^+ A$
- ❑ This way, set of reactions resulting from executing all the tasks do not depend on the task schedule

Task Properties

- ❑ Task $A = (R, W, \text{Update})$ is *deterministic* if for every value $u \in Q_R$ there is a *unique* value $v \in Q_W$ such that $(u, v) \in [\text{Update}]$
- ❑ If all tasks of a component are deterministic, what can we conclude about the component itself?
- ❑ Task $A = (R, W, \text{Update})$ is *input-enabled* if for every value $u \in Q_R$ there exists at least one value $v \in Q_W$ such that $(u, v) \in [\text{Update}]$
- ❑ If all tasks of a component are input-enabled, what can we conclude about the component itself?

Credits

Notes based on Chapter 2 of

[Principles of Cyber-Physical Systems](#)

by Rajeev Alur

MIT Press, 2015